

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics
Honour School of Economics and Management
Honour School of History and Economics

MICROECONOMICS

TRINITY TERM 2019

Wednesday 29 May, 09:30–12:30

Please start the answer to each question on a separate page.

There are 10 questions in this paper.

*Answer **all** questions from Part A and **two** questions from Part B.*

Part A attracts 40% of the marks in total; each question in Part B attracts 30% of the marks.

*In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets;
in Part B, the weight assigned to each part of the multipart question is indicated in square brackets.*

Candidates may use their own calculators.

Do not turn over until told that you may do so.

Part A

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets

1. Consider an economy with one consumer, one firm (owned by the consumer), and two types of good, X and Y .

The consumer owns the endowment of the economy, namely 14 units of good X , and her utility function is $u(x_c, y) = \ln x_c + 2 \ln y$.

The firm can transform good X into good Y ; if it uses an amount x_f of good X then it produces an amount $y = x_f^{2/3}$ of good Y .

- (a) Find the consumer's MRS and the firm's MRT.
- (b) Show that the allocation in which she consumes $x = 6$ and $y = 4$ is efficient.
- (c) Take good Y as the numeraire. What relative price of good X is required for this allocation to be a competitive equilibrium?
Find the firm's profit in this equilibrium, and verify that the consumer's budget constraint (which includes the firm's profit as her income) is satisfied.
- (d) Illustrate the equilibrium in a diagram showing the production possibility frontier of the economy.

[30%]

2. (a) What does it mean for a strategy to be dominant?

If players choose dominant strategies, will they play according to a Nash Equilibrium?

And if they play according to a Nash Equilibrium, will they play dominant strategies?

- (b) For each of the following two-player one-shot games, discuss whether or not you can predict how the players will play.

Game 1. Each player decides between *Green* and *Red* and they simultaneously announce their choices, which result in the following pay-offs:

if they chose the same action then they each receive 5, otherwise the one that chose *Green* receives 6 and the one that chose *Red* receives 4.

Game 2. Each player decides between *Blue* and *Yellow* and they simultaneously announce their choices, which result in the following pay-offs:

if they both chose *Blue* then they each receive 3, whereas if they both chose *Yellow* then they each receive 5; otherwise the one that chose *Blue* receives 4 and the one that chose *Yellow* receives 2.

- (c) Comment on the usefulness of the Nash Equilibrium concept for predicting the outcome of a game.

[20%]

3. (a) Define the certainty equivalent of a lottery. How can it be used to determine whether or not someone should participate in a lottery?

There is a lottery whereby anyone with a ticket will win either £12 or nothing, each with probability $\frac{1}{2}$.

- (b) Janet is an expected-utility maximiser with utility function $u(y) = 2y^{1/2}$ over final wealth y . She has initial wealth of £4 and also a ticket for the lottery.

What is her expected utility?

What is the lowest price that she would be willing to sell the ticket for?

- (c) Sam is also an expected-utility maximiser and has the same utility function as Janet. Sam also has initial wealth of £4 but doesn't have a ticket for the lottery.

What is the highest price that Sam would be willing to pay for a ticket?

- (d) Illustrate your results from part (b) and part (c) in a diagram showing wealth after winning the lottery on the horizontal axis, and wealth after no pay-out from the lottery on the vertical axis.

Which is larger, Janet's lowest selling price or Sam's highest buying price, or are they the same? Explain.

[30%]

4. A risk-neutral principal hires a risk-averse agent to undertake a project. There are two possible effort levels that the agent can exert, $e = 0$ or 1 , and two possible outcomes for the gross profit to the principal, $\pi_L < \pi_H$. If $e = 0$, the probabilities of π_L, π_H are $5/6, 1/6$ respectively; if $e = 1$, the probabilities of π_L, π_H are $5/8, 3/8$ respectively.

The agent's utility from receiving a wage w and exerting effort e is $\sqrt{w} - 5e$, and the agent has a reservation utility of 10.

- (a) When effort is observable (& contractible), what contract should the principal offer the agent if she wants high effort, and why?

Now assume that effort is unobservable: in the contract between the two parties, w cannot depend on e directly, but can depend on the outcome, π_k . Thus a contract will be of the form $\{w_L, w_H\}$, where w_k is the wage paid by the principal to the agent if the gross profit is π_k . Assume that the principal wants to induce high effort.

- (b) If the lower wage, w_L , is 36, find the higher wage, w_H , such that the participation constraint is binding.

Show that, with these values of w_L and w_H , the contract is incentive compatible.

- (c) In words, what is the "agency cost" associated with unobservable effort?

Calculate the expected wage, and hence determine the agency cost in this case.

[20%]

Part B

5. The university's Student Union (SU) has asked you to advise them on the design of a voting system to elect the next SU President. Every student is eligible to vote, and it is likely that there will be many candidates.

Discuss the key trade-offs in voting system design, and whether there is a particular voting system that you would recommend.

6. "Tradeable conscription permits are an efficient way to provide national defence."

Discuss.

7. What criteria should the competition authority use when approving, or prohibiting, a merger?

Under what circumstances are these criteria likely to hold?

(Credit will be given for the use of examples or case studies to illustrate your answer.)

8. There is a lottery such that anyone with a ticket will win one of three prizes: low (£40), medium (£50), or high (£60). Judith is indifferent between buying a ticket and not; so is Hamnet.

Consider three different changes (each to the basic set-up):

- (a) the current wealth of each of them increases by £1000;
- (b) the probabilities of the low and high returns both increase by the same amount;
- (c) the probability of the low return increases and the probability of the high return decreases by the same amount.

As a result of any of these changes, Judith would be willing to buy a ticket, but Hamnet wouldn't.

For each of the three changes, can their responses be reconciled with Judith and Hamnet being expected-utility maximisers with increasing utility functions?

9. "When prospective employees are able to signal their ability, competitive employers are no better off, employees with lower productivity are definitely worse off, and even employees with higher productivity might be worse off."

Do you agree? Does it follow that removing the ability to signal would result in a Pareto improvement?

10. Consider an exchange economy with three sorts of good: apples (A), bananas (B), and cherries (C).

There are 20 traders, 10 of type 1, and 5 each of types 2 and 3, whose initial allocation is as follows:

- type 1 traders each own 10 apples,
- type 2 traders each own 5 bananas,
- type 3 traders each own 20 cherries.

All traders have the same utility function that depends only on their consumption of apples (x_A), bananas (x_B), and cherries (x_C):

$$u(x_A, x_B, x_C) = x_A^{1/2} x_B^{1/4} x_C^{1/4}.$$

Let p_A , p_B , and p_C denote the prices of the three sorts of good, and normalise p_C (the price of cherries) to 1. Suppose that all traders can trade freely taking prices as given.

- (a) Will all types of trader engage in trade? Why?

What is the demand for apples, bananas and cherries for each type of trader?

[30%]

- (b) Calculate the excess demand for apples and bananas in the economy.

Calculate the competitive equilibrium prices.

What is the excess demand for cherries in this competitive equilibrium? Why?

[40%]

Now consider a different allocation (apples, bananas, cherries) for the traders:

- each type 1 trader has $(9, 9/4, 9)$,
- each type 2 trader has $(1, 1/4, 1)$,
- each type 3 trader has $(1, 1/4, 1)$.

- (c) Is this allocation Pareto efficient?

If it is, find the supporting competitive equilibrium prices.

If it isn't, propose a Pareto-improving trade.

[30%]